Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

graphicx,amssymb, amstext, amsmath, epstopdf, booktabs, verbatim, gensymb, geometry, appendix, natbib, lmodern letterpaper

An Example Document Andrew Hobbs 19 Pebruari 2025

cpistuff/cpi

This is a document describing this template.
Key Findings After careful analysis, a number of conclusions have been reached:**This template works okay.** Bu

1 Text Tricks

This section describes some custom CPI commands.

1.1 Orange writing

Use the `callout` command:

By the shores of gitchee gumee
by the shining big sea waters
stood the wigwam of Nokomis
brother of the moon, Nokomis.

1.2 Writing in a box

Use the `frame` command:

'Twas brillig and the slithy toves did gyre and gimble in the wabeall mimsy were the borogroves, and the mome rath

2 Presenting data

This describes more general stuff. If you are experienced with LaTeX, you may already know this.

2.1 Figures

Gambar 1: This is a fat horse

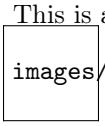


Figure 1 shows a fat horse. You might use a more informative graphic. If you make it in excel, make sure the dimensions are right before you export it, or it will be fuzzy. I do that, and then copy it into paint and save it as a *.png (or *.jpg). For Stata graphics, you can either save them as vector graphics (*.eps) and use `epstopdf` or save them as a normal graphic. The advantage of vector graphics is that you can scale them without fuzziness.

2.2 Tables

Table 1 was produced using `est2tex` to get results directly out of from Stata. The tables were beautified using the `booktabs` package. Both of these things are worth doing, and will save time and enhance appeal.

Tabel 1: Effect of Codes

	Natural Gas		Electricity		Gas & Electricity	
2-7	(1)	(2)	(3)	(4)	(5)	(6)
Log Electric Price	.412***	.410***	-.307***	-.334***	.048	.028
	(.111)	(.053)	(.048)	(.027)	(.055)	(.028)
Log Gas Price	-.314***	-.472***	.132***	.202***	-.073***	-.099***
	(.047)	(.032)	(.013)	(.013)	(.021)	(.014)
Percent Code Units	.470*	.799***	-.001	-.147***	.215	.267***
	(.281)	(.110)	(.099)	(.044)	(.145)	(.051)
Log Heating Degree Days	.697***	.621***	.113***	.146***	.383***	.370***
	(.072)	(.058)	(.024)	(.021)	(.043)	(.027)
Log Cooling Degree Days	.011	.008	.058***	.061***	.015	.015*
	(.017)	(.019)	(.012)	(.010)	(.010)	(.009)
Log Median Income	.089	.116*	.031	.008	.034	.030
	(.119)	(.065)	(.040)	(.028)	(.047)	(.028)
Instrumental Variables	No	Yes	No	Yes	No	Yes

Confidence Level: *90% **95% ***99%